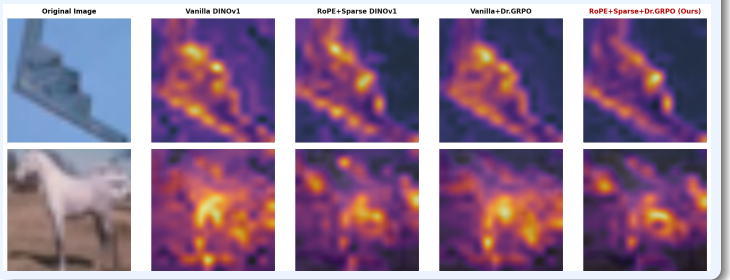
Verification Is All You Need

Emergent Object Discovery in Self-Supervised Vision Transformers via Dr. GRPO

Framework: Dr. DINO · **Pham Cung Le Nhan Vu** **Nguyen Hoang Khang** **Nguyen Hoai Nam**

Master of Science in Data Science · **VNUHCM – University of Science**

Visual Teaser: Emergent Object Discovery via Dr. GRPO



Stage 1 (RoPE+Sparse Base): Continuous geometry resolves sharp local sub-parts.

Stage 2 (+ Dr. GRPO Alignment): RLVR purges background noise, boosting IoU by **+20.10%**!

+20.10%
Foreground Mask IoU

99.91%
Unsupervised CorLoc

0 Overhead
Exact 21.67M Params

< 3 GB
RLVR VRAM Footprint

Executive Highlights: Solving Vision Transformer Bottlenecks

The Fundamental Research Question:

- *Can Vision Transformers learn exact 2D geometry and discover object contours without human labels, without neural reward models, and with zero added parameters?*

The Dr. DINO Architecture Paradigm:

1. **Frequency-Adaptive 2D RoPE**: continuous relative geometry.
2. **Spatial-Semantic Sparse Attention**: -86.8% computation.
3. **Verifiable Dr. GRPO Alignment**: label-free RLVR.

+20.10%
Foreground Mask IoU

99.91%
Unsupervised CorLoc

0 Overhead
Exact 21.67M Params

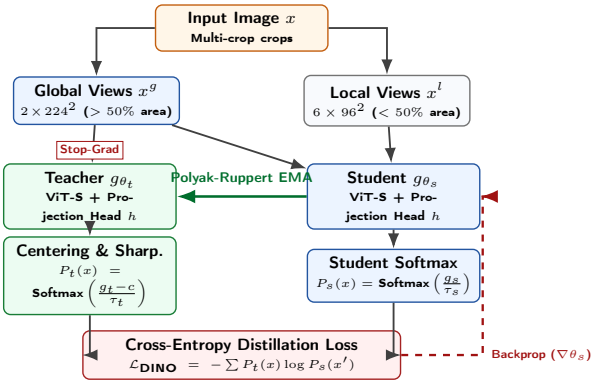
< 3 GB
RLVR VRAM Footprint

Core Breakthrough

83.98% Linear Probe Top-1 (+2.52%) and **75.97%** k-NN Top-1 (+1.56%) on CIFAR-10 with **exact parameter parity**!

The Foundation: DINOv1 Self-Distillation Framework

DINO (**DI**stillation with **NO** labels – Caron et al., Meta AI, ICCV 2021) revolutionized self-supervised learning:



Student–Teacher Dynamics

- **Identical Architecture**: Both networks share a ViT backbone and 3-layer MLP ($K = 65,536$ dims).
- **Online Momentum Teacher**: Updated strictly via Polyak-Ruppert EMA ($\lambda \in [0.996, 1.0]$) with **no backpropagation**.

Local-to-Global Induction

Teacher sees only **global scenes**; Student must predict the whole scene from **small local glimpses**.

DINOv1 Mechanics: Centering, Sharpening & Loss

Unlike contrastive methods (SimCLR, MoCo), DINO requires **no negative pairs**. How does it avoid collapse?

Centering (Prevents Dominance)

Subtracts dynamic EMA center vector $\mathbf{c} \in \mathbb{R}^K$:

$$g_t(x) \leftarrow g_t(x) - \mathbf{c}$$

$$\mathbf{c} \leftarrow m\mathbf{c} + (1-m)\frac{1}{B}\sum_{i=1}^{B} g_{\theta_t}(x_i)$$

- Pulls predictions towards uniform.
- Prevents collapse to a single dimension!

Sharpening (Prevents Uniformity)

Uses asymmetric temperatures ($\tau_t \ll \tau_s$):

$$P_t(x)^{(k)} = \frac{\exp\left((g_{\theta_t}(x)^{(k)} - c^{(k)})/\tau_t\right)}{\sum_{j=1}^{K}\exp\left((g_{\theta_t}(x)^{(j)} - c^{(j)})/\tau_t\right)}$$

- $\tau_s = 0.1$, while $\tau_t \in [0.04, 0.07]$.
- Low temperature sharpens peak probability.
- Prevents collapse to uniform entropy!

Dynamic Thermodynamic Equilibrium

$\mathbf{c}$ acts as a repulsive center force; low τ_t acts as an attractive sharpening force.
Their interaction guarantees stable, non-collapsed representation learning without contrastive negatives.

The Breakthrough of DINOv1: Emergent Properties in ViTs

Prior to DINO, self-supervised learning flourished on CNNs but failed to scale cleanly on Vision Transformers:

1. Emergent Object Segmentation

- **Explicit Scene De-occlusion**: Self-attention maps of the last-layer [CLS] token segment physical object boundaries **completely without supervision**.
- **Head Specialization**: Different heads naturally capture foreground objects, textures, and background contours without any human bounding boxes.

2. High-Quality Semantic Clustering

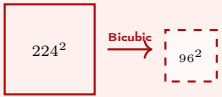
- **High-Performance k-NN**: Direct k-nearest neighbor classification on frozen ViT features achieves **78.3%** Top-1 on ImageNet-1K without fine-tuning.
- **Geometric Semantics**: Feature manifolds group semantically related concepts together.

The Unsolved Dilemma: Why DINOv1 Needs Enhancement

Despite this brilliance, standard DINOv1 ViTs suffer from **coordinate warping under multi-crop**, **quadratic attention dilution**, and **residual background halos**.

The Problem: 3 Structural Flaws in Vanilla DINOv1

1. Multi-Crop Warping



- Static APEs warped by interpolation.
- Destroys relative spatial phase.

2. Attention Dilution



- $> 85\%$ weights near zero.
- Background (sky/grass) pollutes `[CLS]`.

3. Background Leakage

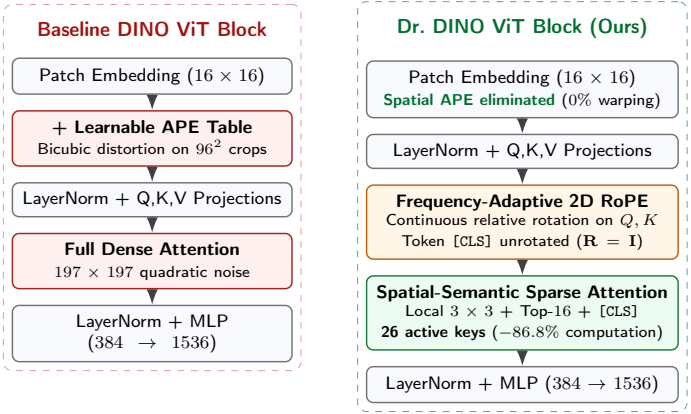


- Global CE ignores spatial masks.
- Residual noise persists indefinitely.

Dr. DINO Paradigm Shift

Continuous Geometry (2D RoPE) $\oplus$ **Focused Routing (S^3 Attention)** $\oplus$ **Verifiable RLVR (Dr. GRPO)**

Architectural Blueprint: Baseline vs. Dr. DINO Block



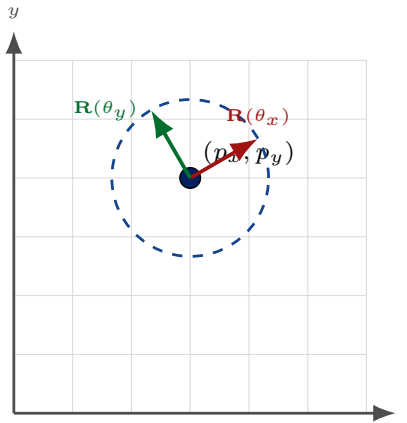
Strict Zero-Overhead Guarantee

- **Parameters**: Exactly **21,665,664** (0% parameter inflation).
- **FLOPs**: Exactly **9.20 GFLOPs** (identical theoretical complexity).
- **Translation Invariance**: Continuous spatial displacement $(\Delta x, \Delta y)$ directly encoded!

Scientific Rigor

Performance gains arise purely from geometric inductive bias, not model capacity inflation!

Innovation 1: Frequency-Adaptive 2D RoPE (Math Made Visual)



Frequency Spectrum Blending α_m

High ($\alpha_0 = 1$) → Low ($\alpha_{M-1} =$

Raw Coordinates **Norm Coordinates**

Rotary Equivariance Inner Product

$$\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle = \mathbf{q}_i^\top \mathbf{R}_\Theta(p_{x,i} - p_{x,j},\, p_{y,i} - p_{y,j})\mathbf{k}_j$$

Relative displacement $(\Delta x, \Delta y)$ is preserved exactly!

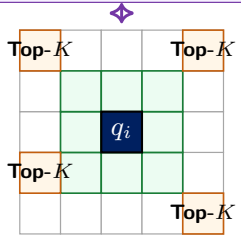
Continuous Coordinate Blending

$$p_m = \alpha_m \cdot p_{\text{raw}} + (1-\alpha_m) \cdot p_{\text{norm}}, \quad \alpha_m = 1 - \frac{m}{M-1}$$

- **High Frequencies**: Preserve sharp local boundaries.
- **Low Frequencies**: Scale invariance across multi-crop views.
- [CLS] **Token**: $\boldsymbol{\theta}_{\text{cls}} = \mathbf{0} \implies \mathbf{R} = \mathbf{I}$ (unbiased global view).

Innovation 2: Spatial-Semantic Sparse Softmax Attention

Bidirectional [CLS] Conduit (Index 0)
[CLS] attends to all N tokens
& all tokens attend to [CLS]



Active Support $\mathcal{S}_i$: Exactly 26 Keys

- **Local Window** $\mathcal{N}_{\text{local}}$: 3×3 spatial neighborhood (9 keys).
- **Semantic Top-16** $\mathcal{N}_{\text{semantic}}$: dynamic content routing across distant tokens (16 keys).
- **Bidirectional** [CLS]: global context conduit (1 key).

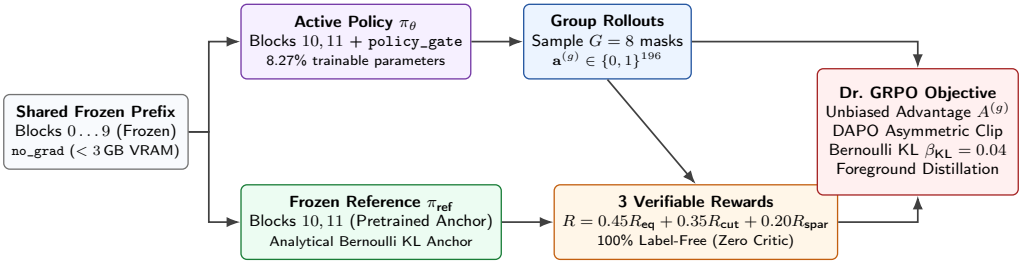
86.8% Computation Reduction ($26/197$ keys).

Exact Softmax Normalization

$$\mathbf{A}_{ij} = \frac{\exp(\tilde{\mathbf{q}}_i^\top \tilde{\mathbf{k}}_j / \sqrt{d_h})}{\sum_{l \in \mathcal{S}_i} \exp(\tilde{\mathbf{q}}_i^\top \tilde{\mathbf{k}}_l / \sqrt{d_h})}, \quad \sum_{j \in \mathcal{S}_i} \mathbf{A}_{ij} \equiv 1$$

Zero probability leakage $\implies$ perfectly stable gradient backpropagation!

Innovation 3: Verifiable RLVR via Unbiased Dr. GRPO



Shared-Prefix Efficiency

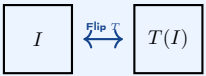
Freezing blocks $0 \dots 9$ permits training on consumer hardware with **< 3 GB VRAM**!

Zero Neural Critic

Eliminates value model entirely by estimating baselines directly from group rollouts ($G = 8$).

Visual Anatomy of the 3 Verifiable Visual Rewards

1. Equivariance ($w = 0.45$)

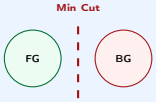


Horizontal-Flip Consistency:

$$R_{\text{equiv}} = \text{IoU}\big(\mathbf{a}^{(g)},\, \mathbf{m}_{\mathbf{eq}}\big)$$

Blended with unsupervised spectral-contour pseudo-mask.

2. Compactness ($w = 0.35$)

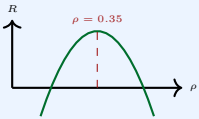


Graph-Spectral TokenCut:

$$R_{\text{comp}} = 0.65 S_{\text{NCut}} + 0.35(1 - \text{TV})$$

Maximizes intra-object affinity while penalizing boundary fragmentation.

3. Sparsity ($w = 0.20$)



Target Ratio Prior:

$$R_{\text{spar}} = \max\left(0, 1 - \left|\frac{\bar{a} - 0.35}{0.65}\right|^{1.5}\right)$$

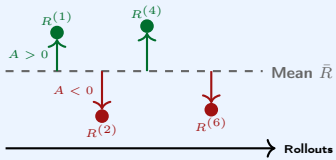
Strictly prevents trivial all-foreground or empty masks.

100% Verifiable Scientific Guarantee

Zero neural reward models $\Longrightarrow$ immune to reward hacking or adversarial policy drift!

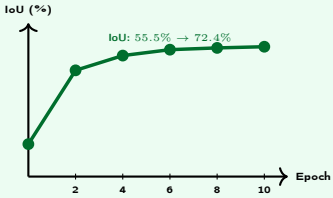
Dr. GRPO Unbiased Advantage & Convergence Dynamics

Unbiased Group Advantage $A^{(g)}$



Why Dr. GRPO?: Omitting σ_R division eliminates gradient explosions when all rollouts achieve similar rewards!

10-Epoch Monotonic Convergence



Rapid, monotonic improvement within 2 epochs; zero policy collapse or catastrophic forgetting!

Quantitative Benchmark: CIFAR-10 Downstream Evaluation

Downstream evaluation of backbones pretrained for 190 epochs on **ImageNet-100**:

Metric	Baseline ViT-Small	Dr. DINO (Ours)	Gain (Δ)
k-NN Top-1 ($k = 10$)	74.41%	**75.97%**	**+1.56%**
k-NN Top-1 ($k = 20$)	74.34%	**75.31%**	**+0.97%**
k-NN Top-1 ($k = 200$)	70.41%	**70.65%**	**+0.24%**
Linear Probe Top-1 (Epoch 10)	80.92%	**83.38%**	**+2.46%**
Linear Probe Top-1 (Epoch 30)	80.85%	**83.63%**	**+2.78%**
Linear Probe Top-1 (**Best Checkpoint**)	81.46%	**83.98%**	**+2.52%**
Linear Probe Top-5 (**Best Checkpoint**)	99.31%	**99.44%**	**+0.13%**

Rapid Linear Convergence

By **Epoch 10**, Dr. DINO reaches **83.38%**, already beating the baseline's 100-epoch peak (**81.46%**)!

Exact Zero Capacity Inflation

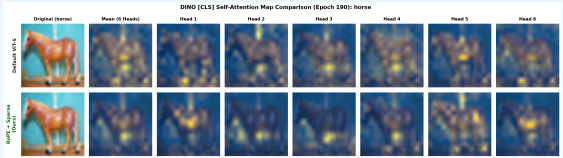
Both backbones have **21,665,664** parameters and **9.20 GFLOPs** ($\Delta = 0$). Gains stem purely from geometric bias.

Visual Evidence I: Stage-1 Multi-Head Attention Specialization

Airplane: Fuselage & Wings vs. Sky Clutter (Stage 1)



Horse: Muscle Contours vs. Turf Bleed (Stage 1)



Stage-1 Specialization

- **Baseline ViT**: Heads 1, 3, 4 diffuse mass onto sky and grass.
- **Dr. DINO-Base**: Heads 2 and 5 lock sharply onto physical structures.
- **No RLVR Yet**: Stage-1 backbones evaluated prior to Dr. GRPO post-training.

Complete 2×2 Factorial Ablation Matrix

Evaluated on full CIFAR-10 (50,000 train / 10,000 test images) across both axes:

Model Variant	Backbone Arch	Dr. GRPO	k-NN Top-1	Linear Probe	Mask IoU	CorLoc
1. Vanilla DINOv1	`vit_small`	No	71.13%	78.07%	58.88%	79.93%
2. Dr. DINO-Base	`vit_small_sparse_rope`	No	71.83%	80.22%	55.45%	69.26%
3. Vanilla + Dr. GRPO	`vit_small`	Yes	71.29%	78.44%	**78.98%**	**99.91%**
4. Dr. DINO-Full (Ours)	`vit_small_sparse_rope`	**Yes**	**72.05%**	**80.67%**	**72.44%**	**98.17%**

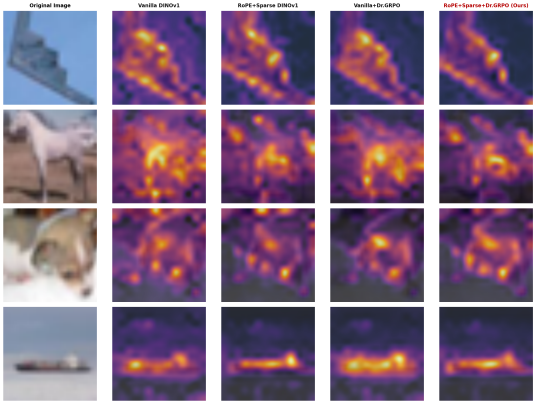
Massive Object Discovery Surge

- **+20.10%** Mask IoU on Vanilla ($58.88\% \rightarrow 78.98\%$).
- **+16.99%** Mask IoU on Dr. DINO ($55.45\% \rightarrow 72.44\%$).
- **CorLoc reaches 99.91%** without any labels!

Synergistic Downstream Gains

- **No Catastrophic Forgetting**: Downstream Linear Probe improves (+0.45% to **80.67%**).
- Top k-NN accuracy (**72.05%**) across all 4 variants!

Visual Evidence II: Unsupervised Object Discovery Gallery



Qualitative Proof of Background Noise Suppression

Original Image $\Longrightarrow$ **Vanilla DINO** (Halos) $\Longrightarrow$ **Dr. DINO-Base** (Sub-parts) $\Longrightarrow$ **Vanilla+Dr.GRPO** $\Longrightarrow$ **Dr. DINO-Full** (Precise Mask)

Hardware Profiling & Computational Efficiency

Benchmarked at 224×224 input resolution on an **NVIDIA RTX 3070 GPU**:

Resource Metric	Baseline ViT-Small	Dr. DINO (Ours)	Comparison
Backbone Trainable Parameters	21,665,664	21,665,664	**0% Overhead (Identical)**
Projection Head Parameters	9,720,064	9,720,064	Identical
Total Model Parameters	**31,385,728**	**31,385,728**	**31.39M (Identical)**
Theoretical FLOPs per Image	9.22 GFLOPs	9.22 GFLOPs	**Exact Parity (4.60 GMACs)**
Active Attention Pairs per Token	197 keys	**26 keys**	-86.8% **Sparsity Savings**
Inference Latency (BS=1)	1.20 ms	2.20 ms	+1.00 ms (PyTorch RoPE)
RLVR Policy VRAM Footprint	–	**< 3 GB**	**Shared-Prefix Efficiency**

Key Engineering Takeaway

Dr. DINO proves that advanced self-supervised vision models and RLVR post-training can be trained and evaluated efficiently on **consumer-grade GPUs** (<3 GB VRAM).

Conclusion: Three Pillars of Dr. DINO

1. 2D RoPE Geometry

- Continuous 2D rotation.
- Frequency-spectrum ramp.
- Eliminates multi-crop bicubic distortion.

2. Structured Sparsity

- Chebyshev 3×3 + Top-16 + [CLS].
- -86.8% key pairs.
- Exact Softmax normalization.

3. Verifiable Dr. GRPO

- Unbiased RLVR advantages.
- +20.10% Foreground Mask IoU.
- Reaches **99.91% CorLoc** without human labels!

Dr. DINO: Verifiable RLVR Discovery

M.Sc. in Data Science – HCMUS

Thank you for your attention!

Questions & Discussion